

of Employment and Wages (QCEW), which is a near universe of quarterly employment (but lacks information on hourly wages). [Online Appendix F](#) explains the QCEW benchmarking in detail. As we discuss below, the QCEW benchmarking has little effect on our point estimates but substantially increases their statistical precision.

Our estimation of the change in jobs paying below and above a new minimum wage requires us to specify minimum wage-increasing events. For state-level minimum wage levels, we use the quarterly maximum of the state-level daily minimum wage series described in [Vaghul and Zipperer \(2016\)](#).¹⁴ For the 138 minimum wage events, on average, 8.6% of workers were below the new minimum wage in the year before these events and the mean real minimum wage increase was 10.1%.¹⁵

One concern when using \$0.25 bins and CPS data is that some of the bins may be sparse with very few or no workers. However, we stress that our employment estimate is based on the sum of employment changes in 36 cells covering a \$9 range [$MW - \4, $MW + \$4$], summed over at least 4 quarters (typically 20 quarters). As a result, small or zero employment in particular cells is not a major concern. In each state, there are, on average, around seven workers each quarter in each of the \$0.25 bins between \$5 and \$15/hour in our sample.¹⁶ Since the coefficients for our event dummies are estimated at a \$1-bin-year-state level, on average, for these we use around 112 individual-level observations per event. Moreover, when we assess the total employment effects, we calculate the sum of the \$1-bin estimates between \$4 below and \$4 above the minimum wage, and we consider five-year averages. This implies that, on average, we use approximately 5,040 individual worker observations per event. This is a well-sized sample that allows a reliable estimate of the true counts of employment for each event. Consistent with this point, we note

14. The minimum wage series is available at <https://github.com/benzipperer/historicalminwage/releases>.

15. All minimum wage increases including our events are shown in [Online Appendix Figure A.1](#)

16. Overall, we have 847,314 wage-bin-state-period observations, which we obtained from 4,694,104 individual-level observations, producing a count of 5.5 workers per \$0.25 bin. However, the count per bin is higher in the \$5-to-\$15/hour range because the upper-tail wage bins are more sparse. The \$5-to-\$15/hour range is the relevant one because it contains the [$MW - \$4$, $MW + \$4$] windows for all of our events.

again that results from our approach are very similar to those from a simpler method that uses state-by-quarter data and where the outcomes are the (per capita) number of jobs or total wage bill under \$15/hour.

Another potential concern with the data is that misreporting of wages in the CPS may bias our estimates. If reported wages contain some measurement error, some workers earning above the minimum wage will appear to earn below it, which could attenuate the estimate for Δb . However, this does not affect the consistency of the estimate for $\Delta a + \Delta b$ as long as the minimum wage only affects reported wages below $\bar{W}$. The reason is straightforward. Assume that 1% of the workforce mistakenly report earning below the new minimum wage in the post-treatment period. This would lead our estimate of the missing jobs to be too small in magnitude: $\hat{\Delta b} = \Delta b + 0.01$. However, this misreporting would also lead to an equal reduction in the number of excess jobs above, producing the estimate $\hat{\Delta a} = \Delta a - 0.01$; this will be true as long as these misreported workers are coming from the range $[MW, \bar{W})$, which is likely to be satisfied for a wide variety of classical and non-classical measurement error processes where the support of the measurement error is contained in $[MW - \bar{W}, \bar{W} - MW]$. Therefore, the employment estimate $\hat{\Delta a} + \hat{\Delta b}$ is likely to be unaffected by measurement error in reported wages. We also directly assess how misreported wages in the CPS may affect our results in [Online Appendix E](#), where we compare the CPS hourly wage distribution to microaggregated administrative data on hourly wages from three U.S. states that collect this information. Reassuringly, the evolution of the number of jobs paying below the minimum wage, and the number of jobs paying up to \$5 above the minimum wage in the CPS data from these three states match quite well with their counterparts using administrative data.¹⁷

17. In [Online Appendix F](#), we also structurally estimate a model of measurement error in reported wages proposed by [Autor, Manning, and Smith \(2016\)](#), and show that the contribution of misreporting error to the overall variance in wages in the CPS and in administrative data on hourly wages from three states are very similar. Furthermore, we semiparametrically deconvolve the CPS wage distribution using the estimated measurement error model and show that our estimates using this measurement error-corrected distribution are very similar to the baseline estimates ([Online Appendix Table F.3](#)). In [Online Appendix C](#) we implement our approach using administrative data from Washington and find estimates to be similar when using the CPS. Although each piece of evidence has limitations, together they suggest that our employment and wage results are not likely to be

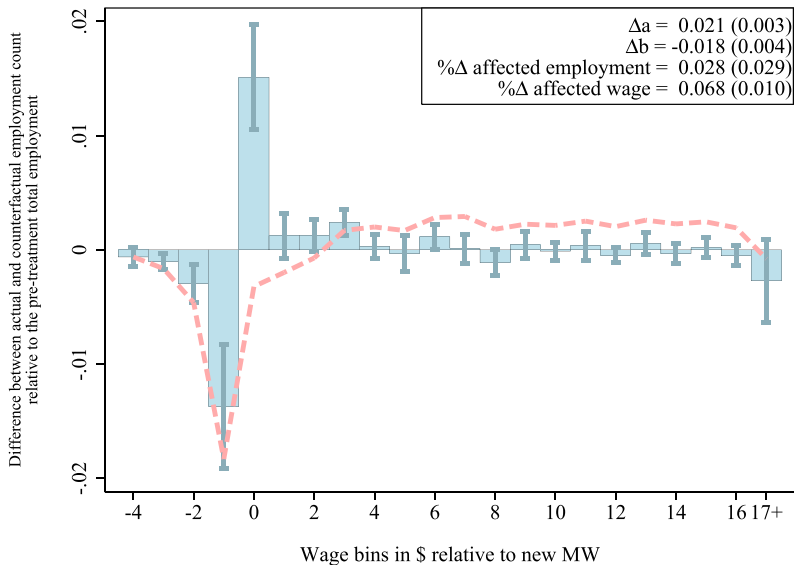


FIGURE II

Impact of Minimum Wages on the Wage Distribution

changes are normalized to pretreatment total employment in the state. Several points should be noted.

First, there is a clear and significant drop in the number of jobs below the new minimum wage, amounting to 1.8% (std. err. 0.4%) of the total pretreatment employment.¹⁸ Around $\frac{3}{4}$ of this reduction occurs in the \$1 wage bin just under the new minimum. Second, there is also a clear and significant increase in jobs just at the new minimum wage (at the \$0 wage bin). Third, there is also a statistically significant increase in employment in the wage bin \$3 above the new minimum and modest, statistically insignificant increases in the \$1 and \$2 bins. This pattern of employment changes is consistent with limited wage spillovers resulting from the minimum wage increase, as suggested in [Autor, Manning, and Smith \(2016\)](#).¹⁹ Fourth, the excess jobs between the new minimum and \$4 above it represents 2.1% (std. err. 0.3%) of the total pretreatment employment. Fifth, the employment changes in the upper-tail wage bins, from \$5 above the minimum wage to \$17 or more (the final bin), are all small and statistically insignificant—individually and cumulatively as shown by the red line, which represents the running sum of employment changes. Finally, it is worth emphasizing that the drop in employment just below the new minimum, the equal-sized increase just above it, and the lack of employment change in the upper tail is exactly what we expect if employers are complying with the law and adjusting wages but not employment.

We estimate the employment change by adding the missing jobs below and excess jobs above the minimum wage: $\Delta a + \Delta b$. We divide this change by the jobs below the new minimum wage ($\bar{b}_{-1} = 8.6\%$) to obtain a change in the affected employment of

18. The discrepancy between the actual number of jobs below the new minimum, which is 8.6% of total pretreatment employment on average, and the change in the number of jobs below it, which is 1.8% on average, can be explained by the following factors. First, some of the jobs below the minimum wage (e.g., tipped workers) are exempted from the minimum wage in most states. Second, there are often multiple changes in the minimum wage in a relatively short period. In these cases, the cumulative effect of the various treatments should be considered: when we adjust for this we find the change in the number of jobs below the minimum rises in magnitude from 1.8% to 2.5%. Third, there is some wage growth even in the absence of a minimum wage increase, and our event study design controls for these changes.

19. The \$3 above the minimum wage is around the 23rd percentile of the wage distribution on average. [Autor, Manning, and Smith \(2016\)](#) finds the wage spillovers are effectively zero at around the 25th percentile.